



SAINT PIERRE
LEARNING RESOURCES

COMPUTER SCIENCE

Introduction to Programming

Python Decisions

College • Introductory Computer Science

INSIDE THIS RESOURCE

Teacher guide and student practice

Assessment and answer key

Differentiation and accessibility supports

PRINTABLE PDF | SINGLE-TEACHER LICENSE

saintpierreresources.com



Teacher Guide

Level: College • Introductory Computer Science

Standard / objective: Course objective — Trace variables and conditionals, write Boolean expressions, and create small input-process-output programs in Python.

Source / course note: Course-level objective; no accreditation or universal equivalency claimed.

Before this lesson: Recognize variables, numbers, and strings.

Model and discuss: Trace $x=3$. The condition $x>5$ is False, so the else branch doubles x to 6. Emphasize that only one branch runs and indentation defines the block.

Watch for: Use $==$ to compare and $=$ to assign. `elif` and `else` belong on separate lines with correctly indented bodies.

Suggested sequence: 5-minute launch, 10-minute model, 20-30 minutes of practice, then a 5-minute exit ticket. Use separate paper for extended writing.

Learning supports

AIG: Add a boundary case and explain whether the threshold should use $>$ or $>=$.

504: Offer chunked tasks, reduced copying, extra time, and breaks as specified in the learner's plan.

EC: Use a trace table with columns for line, condition, and variable value.

ESL: Rehearse: The condition is ___, so the program executes ___ and outputs ___.

Adapt these suggestions to the current learning plan. Print on US Letter paper at actual size; keep the key separate. One teacher may use the resource with their own students; no public posting or resale.

Student Directions & Practice

Trace and write short Python programs using if, elif, else, comparison, and Boolean operators.

Directions: Complete every item. Show work or cite evidence. For open responses, answer in complete sentences.

1. What prints? `x=7; print(x+3)`
2. Evaluate: `5 < 8` and `2 == 2`
3. Write a condition for age at least 18.
4. What prints? `n=4; print("even" if n%2==0 else "odd")`
5. Find the bug: `if score = 90:`
6. Write code that prints “warm” when `temp>75`.
7. Why should input converted with `int()` be validated?
8. Trace this rule: start with `x=3`; if `x` is greater than 5, add 2; otherwise multiply `x` by 2. What is `x`?



Assessment / Exit Ticket

Work independently. Show your reasoning. Use extra paper if needed.

1. If score is 75, what does this rule print: `print("pass" if score >= 70 else "retry")`? Explain.

2. Write a Boolean condition that is True when age is at least 18 and has_id is True.

Reflection: What helped you decide? What do you want to practice next?

Score each task: 2 = accurate with clear reasoning; 1 = partly correct or missing reasoning; 0 = incorrect or no evidence. The reflection is not scored.

Answer Key & Teaching Guidance

1. 10
2. True
3. `age >= 18`
4. even
5. Use `==` for comparison.
6. `if temp > 75: print("warm")`
7. Non-numeric input can raise an error; validation improves reliability.
8. 6

Exit ticket guidance

1. pass, because `75 >= 70` is True.
2. `age >= 18 and has_id` (or `age >= 18 and has_id == True`).

Next teaching move: Use the worked model again if the concept is unclear. If the answer is correct but reasoning is incomplete, ask for an explanation. Offer the extension once both tasks are secure. Scores guide instruction; they are not standardized mastery cutoffs.